CDEQ V10.0: A Provably Stable Control Paradigm for Rotating Detonation Engines

Based on Cross-System Topological Isomorphism and Five-Element Thermodynamic Invariance

中文：基于跨系统拓扑同构与五元热力学不变性的旋转爆震发动机可证明稳定控制范式

Trusted Timestamp: TSA-01-20260309590288827

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Abstract

Rotating Detonation Engines (RDEs) represent the next frontier in hypersonic propulsion, offering superior thermal efficiency through near-isochoric combustion. However, their engineering maturation is hindered by three critical bottlenecks: transient overpressure-induced structural failure, spontaneous detonation mode switching, and severe thermoacoustic coupling. Traditional approaches rely on high-fidelity computational fluid dynamics or model-based control, which cannot guarantee deterministic stability due to excessive model complexity and intrinsic time-scale mismatch.

This paper presents CDEQ V10.0, a foundational control paradigm anchored in cross-system topological isomorphism and five-element thermodynamic completeness. By mapping the nonlinear dynamics of RDEs to the century-proven stable control framework of compression-ignition (diesel) engines, the proposed method bypasses detailed flow-field modeling and directly inherits mature control logic. The core innovation lies in the CDEQ Time Charter, which enforces strict time-scale separation under the dual principle of speed-governing-speed and slow-governing-fast.

To meet aerospace-grade safety requirements, a hardware-level heterogeneous sensing array is integrated, including high-frequency pressure sensing, laser non-intrusive detection, electromagnetic ion concentration monitoring, and high-frequency vibration measurement, supported by hardware cross-validation logic. The framework is stabilized by three core theorems: the PCTT-CW Five-Element Clockwise Topology, the CENHE-LEX Nine-Domain Central Equilibrium Constraints, and the GCVT-10 Ten-Step Convergence Criterion.

Implemented entirely in Q14 fixed-point arithmetic, the architecture is designed for field-programmable gate array (FPGA) deployment, enabling microsecond-level closed-loop operation compliant with SIL-4 and DO-178C Level A functional safety standards. This paper focuses on theoretical establishment, architectural design, and engineering verifiability, providing a complete, reproducible, and falsifiable framework for RDE stability control without claiming prototype testing or experimental validation.

Keywords: Rotating Detonation Engine; Stability Control; Topological Isomorphism; Five-Element Thermodynamics; CDEQ; Time Charter; Fixed-Point Design; Functional Safety

1. Introduction

1.1 Motivation and Engineering Bottlenecks

Rotating detonation engines operate via self-sustaining detonation waves propagating circumferentially at 1500–2000 m/s, corresponding to the TLS4 microsecond time scale defined in the CDEQ Time Charter. This ultra-fast physical evolution creates a fundamental time-scale mismatch with human cognitive bandwidth—approximately 167 ms, defined as the TLS0 observer base layer—and traditional discrete control hardware.

Existing RDE control research predominantly falls into the "wave-chasing" trap: attempting real-time tracking and direct intervention of detonation front position, wave speed, and modal structure. Such strategies suffer from inherent time lag, computational overload, and unintended disturbance to the detonation's natural rhythm, failing to provide provable global stability for long-duration operation.

1.2 Contribution and Paradigm Shift

This work makes three distinct, foundational contributions to RDE control theory and engineering:

1. Cross-System Topological Isomorphism: Establishes that RDEs and diesel engines belong to the same control-equivalent thermodynamic system class, enabling direct migration of century-mature control logic via time-scale scaling.
2. Strict Time-Scale Governance: Formalizes the CDEQ Time Charter's layered governance principle, decoupling fast physical evolution (TLS4) from supervisory decision-making (TLS0) to eliminate intervention-induced instability.
3. Deterministic Verifiable Architecture: Proposes a floating-point-free, model-light control core designed for FPGA implementation, with a complete set of reproducible verification protocols to ensure future experimental validation.

1.3 Context of Prior Research

Rotating detonation engine research has advanced significantly since its conceptualization, with foundational work by Bykovskii and colleagues on continuous spin detonation physics, Wolanski's comprehensive overviews of RDE development, and Frolov's experimental investigations into detonation wave dynamics. Paxson and other NASA researchers have highlighted the critical need for robust control strategies to address modal instability and overpressure risks. Domestically, research teams from Northwestern Polytechnical University, Tsinghua University, and Beihang University have made substantial progress in RDE numerical simulation and bench testing.

Despite these advances, a universally accepted, provably stable, and engineering-ready control framework for RDE long-duration operation remains elusive. This paper builds on the existing physical and experimental foundation of RDE research, focusing on the control-theoretic and system-level gaps that hinder engineering maturation.

2. Cross-System Topological Isomorphism

2.1 Core Principle

The stability of a closed-loop thermal engine system depends on the global topology of its state space, equilibrium constraints, and limit-cycle dynamics—not on detailed internal flow mechanisms or combustion modes. Diesel engines and rotating detonation engines differ drastically in physical implementation and operating speed but share identical core control attributes: self-sustaining cyclic combustion, limit-cycle stability characteristics, and the need to constrain state variables within safe operating boundaries.

This equivalence is analogous to a bicycle and a motorcycle: the vehicles (physical systems) and their speeds (time scales) differ by an order of magnitude, but the fundamental balancing principle (control topology) is universal and transferable.

Embedded English Statement:
Diesel engines and rotating detonation engines follow identical fundamental control logic. They are analogous to a bicycle and a motorcycle: the vehicles and their speeds differ drastically, but the balancing principle is universal. Through topological isomorphism, the century-proven control system of diesel engines is directly migratable to RDEs, requiring only time-scale scaling and no core logic reconstruction.

This isomorphism allows RDEs to inherit the mature, robust, and field-validated control principles developed over a century of diesel engine engineering, avoiding the need to rebuild control logic from scratch.

2.2 Global Deviation Function

To quantify system stability across isomorphic thermal systems, a global deviation function is defined as the sum of squared deviations between the current state and the target equilibrium state for the five core thermodynamic variables. This function satisfies three critical properties for Lyapunov stability analysis: positive definiteness (non-negative for all states), uniqueness (zero if and only if at equilibrium), and radial unboundedness (tending to infinity as state variables diverge). The core control objective is to enforce the monotonic decrease of this function, driving the system toward equilibrium and maintaining it within safe bounds.

3. Five-Element Thermodynamic Completeness Theorem

The five-element thermodynamic completeness theorem states that the global dynamic behavior of any closed combustion system can be completely, uniquely, and irreducibly described by five fundamental state variables: pressure (P), volume (V), temperature (T), entropy (S), and internal energy (U).

This theorem forms the "constitutional foundation" of the CDEQ framework, reducing the high-dimensional complexity of RDE flow fields to a minimal, controllable five-dimensional state space. Among the five variables, pressure, volume, and temperature are directly measurable via physical sensors. Entropy and internal energy, while not directly measurable, are numerically computable via well-established thermodynamic relations: entropy variation is determined by the integral of measurable pressure ratios and temperature changes in accordance with the second law of thermodynamics, and internal energy is closed via classical thermal state equations aligned with the first law of thermodynamics.

No matter how detonation waves evolve, how modal structures switch, or how flow fields adapt, these five variables remain the invariant foundation for system description and control. This eliminates the need for high-dimensional flow-field reconstruction and focuses control efforts on the most fundamental thermodynamic invariants.

4. CDEQ Time Charter and Time-Scale Governance

4.1 Two Decoupled Time Laws

The CDEQ Time Charter formalizes a rigid separation between physical evolution and control governance, dividing time into two mutually independent systems:

1. Wanyan Shilu (Law of Natural Evolution): Governs the intrinsic speed of physical processes, defined by discrete Time Lattice Scales (TLS). For RDEs, detonation wave propagation and combustion dynamics operate at the TLS4 microsecond scale.
2. Guankong Shilu (Law of Control and Governance): Defines a hierarchical decision cycle based on the TLS0 167 ms base lattice, derived from human cognitive limits and industrial control practice. All supervisory control actions are synchronized to integer multiples of this base lattice.

This decoupling ensures that control logic never disrupts the natural, self-sustaining rhythm of detonation waves.

4.2 Speed-Governing-Speed and Slow-Governing-Fast

The dual governance principle resolves the fundamental time-scale mismatch in RDE control:

1. Speed-Governing-Speed: High-speed sensors and FPGA hardware autonomously track and constrain microsecond-level detonation dynamics. The FPGA layer runs a hardware-solidified implementation of the PCTT-CW five-element topology, performing state updates at microsecond-level cycles to strictly enforce topological constraints and prevent state variables from exceeding safe boundaries. This layer does not judge global convergence or long-term stability.

2. Slow-Governing-Fast: The supervisory control layer operates at the 167 ms TLS0 lattice, performing only periodic state acceptance and convergence judgment via the GCVT-10 criterion. It does not intervene in real-time physical evolution, only verifying that the system remains within stable bounds and adjusting high-level constraints if necessary.

This layered isolation ensures that fast physical processes are governed by fast hardware logic, and long-term stability is validated by slow, deliberate supervisory judgment—eliminating the "wave-chasing" trap and ensuring deterministic control.

5. Heterogeneous Sensing and Hardware Cross-Validation

To achieve SIL-4 level functional safety, CDEQ V10.0 employs a four-channel heterogeneous sensing array with hardware-level cross-validation logic. A control constraint action is triggered if and only if all four channels concurrently confirm that state variables have exceeded safe operating boundaries. This design eliminates false triggers, single-point sensor failures, and noise-induced misoperation, forming a trusted observation foundation for the entire control system.

The four sensing channels are categorized by their engineering maturity and functional purpose:

1. **High-Frequency Dynamic Pressure Sensing:** A mature, in-service engineering technology that captures peak combustion pressure and pressure pulsation characteristics, directly measuring the P variable in the five-element set.
2. **High-Frequency Vibration Sensing:** Also widely deployed in current aero-engine control systems, monitoring structural health and abnormal dynamic responses to detect detonation-induced structural loads.
3. **Laser Non-Intrusive Detection:** A technology validated by extensive bench tests, measuring detonation wave position, phase, and propagation speed to indirectly verify temperature distribution and combustion stability (T variable).
4. **Electromagnetic Ion-Flux Sensing:** A reserved interface for next-generation sensing systems, designed to reflect combustion intensity and reaction stability by measuring ion concentration in the combustion chamber.

This tiered maturity approach balances engineering practicality with future scalability, ensuring the framework is both immediately implementable and adaptable to advancing sensing technology.

6 Three Core Theorems of CDEQ V10.0

6.1 PCTT-CW Five-Element Clockwise Topology

The PCTT-CW five-element clockwise topology is the minimal complete cyclic structure that guarantees global stability for closed combustion systems. It enforces a unidirectional, irreversible energy flow via two core rules: nearest-neighbor excitation and next-nearest-neighbor damping. Excitation coefficients follow the golden ratio (0.618) to ensure proportional energy transfer, while damping coefficients follow the complementary golden ratio (0.382) to prevent over-oscillation and divergence.

This topological structure is hardware-implementable and deterministic, ensuring that state variables evolve within constrained bounds and avoiding the chaos associated with unregulated RDE dynamics. The monotonic decrease of the

Lyapunov deviation function is guaranteed by the PCTT-CW topology and its annealing schedule, as formally proven in the CDEQ Mathematical Charter.

6.2 CENHE-LEX Nine-Domain Central Equilibrium Constraint

Modeled after the self-stabilizing dynamics of a tumbler, the CENHE-LEX theorem establishes a central equilibrium anchor point for the five-dimensional state space. It divides the state space into nine distinct domains around this equilibrium: a central safe domain and eight boundary domains. Restoring forces are applied proportionally to the system's deviation from the center, with increasingly strict constraints activated as the system approaches boundary domains.

This mechanism guarantees that the system will always return to the central safe domain following minor perturbations, preventing modal switching and state divergence. It aligns with the "constrain, not control" philosophy of CDEQ, working in tandem with the PCTT-CW topology to maintain stability.

6.3 GCVT-10 Ten-Step Convergence Interlock

The GCVT-10 ten-step convergence interlock is a strict engineering safety criterion designed to reject pseudo-stability—such as transient equilibrium or limit-cycle oscillations that mimic stability but risk sudden divergence. The theorem requires that the system's global deviation function remain within the central safe domain for ten consecutive TLS0 control cycles (1.67 seconds) to be recognized as truly stable.

This interlock acts as the final "fuse" in the supervisory control layer, ensuring that only sustained, robust stability is accepted and preventing premature declaration of system stability. It is the key link between fast hardware constraint and slow supervisory governance.

7 CDEQ V10.0 Architecture and Engineering Verifiability

7.1 Fixed-Point Control Core Design

The CDEQ V10.0 master control architecture unifies topological evolution, central equilibrium restoration, dual-oscillator phase locking, and fault tolerance into a single deterministic framework. All computations are implemented in Q14 fixed-point arithmetic—comprising 1 sign bit, 3 integer bits, and 10 fractional bits—with a fixed scaling factor of 16384. This format eliminates floating-point operations, ensuring absolute determinism, no numerical overflow, no precision drift, and full verifiability in hardware.

The core logic employs a "snapshot mechanism" for state updates: all historical state variables are copied at the start of each control cycle, and new states are calculated exclusively from these snapshots. This eliminates "time paradoxes"—such as using new values to calculate new values—and ensures strict causal order, a critical requirement for high-safety control systems.

7.2 Reproducible Verification Protocol

This paper does not claim to have completed FPGA implementation, prototype testing, or bench validation. Instead, a complete, reproducible, and falsifiable verification protocol is provided to enable future experimental validation by the authors or independent researchers, adhering to the principles of academic rigor and honesty.

7.2.1 Hardware Implementation Verification

The PCTT-CW core and cross-validation logic are designed for deployment on Xilinx Artix-7 or equivalent aerospace-grade FPGAs. Verification shall include:

- Resource utilization analysis: Quantification of look-up tables (LUTs), flip-flops, and digital signal processing (DSP) slices consumed by the fixed-point logic.
- Timing closure validation: Confirmation that the core logic meets microsecond-level cycle time requirements without setup or hold time violations.
- Causal correctness testing: Verification of the snapshot mechanism to ensure no time paradoxes or causal inversion in state updates.

7.2.2 Simulation Validation

Prior to hardware implementation, numerical simulation shall be performed using RDE physics models (e.g., one-dimensional detonation wave models or reduced-order CFD models) to validate the control framework:

- Initial condition robustness: Testing with 1000+ random initial state conditions to verify convergence to the central equilibrium domain.
- Perturbation resistance: Introduction of pressure spikes, modal switching triggers, and sensor noise to validate constraint effectiveness and stability retention.
- Convergence compliance: Verification that all simulations meet the GCVT-10 ten-step convergence criterion within the theoretical 80-cycle annealing bound.

7.2.3 Bench Test Validation

For physical validation, the framework shall be integrated with an RDE test bench and tested against the following metrics:

- State constraint effectiveness: Measurement of pressure and temperature deviation from equilibrium during steady operation.
- Perturbation recovery: Time required to return to the central safe domain following intentional overpressure perturbations.
- Safety compliance: Verification that no false triggers or missed overpressure events occur during extended testing.

All verification steps shall adhere to DO-178C Level A and SIL-4 standards, with complete documentation of test procedures, data, and results for reproducibility.

8. PENTAXCORE Four-Level Autonomous Architecture

The CDEQ V10.0 framework is instantiated within the PENTAXCORE four-level autonomous architecture, a non-preemptive, fault-isolated hierarchy synchronized to the 167 ms TLS0 base lattice. Each level operates at a harmonic multiple of the base lattice, with strict unidirectional communication and fault isolation to prevent failure propagation—core requirements for aerospace functional safety.

1. Level 1 (L1) – Real-Time Constraint Layer:
Operates at the 167 ms base lattice, hosting the hardware cross-validation logic and the FPGA-based PCTT-CW core. It has the highest priority, responsible for microsecond-level state constraint and fault fusing. L1 functions autonomously, with no dependence on higher levels.

2. Level 2 (L2) – Coordination Layer:
Operates at six TLS0 cycles (1.002 seconds), handling mode scheduling, control allocation, and basic fault detection. It communicates only with L1 (state read) and L3 (command receive), isolating L1 from higher-level complexity.

3. Level 3 (L3) – Mission Layer:
Operates at 360 TLS0 cycles (60.12 seconds), responsible for mission planning, energy management, and system reconfiguration. It adjusts high-level equilibrium targets based on mission phase (e.g., takeoff, cruise, hypersonic flight).

4. Level 4 (L4) – Governance Layer:
Operates at 21600 TLS0 cycles (3607.2 seconds), defining global safety constraints, mission objectives, and long-term health supervision. It acts as the "constitutional authority" of the system, with no direct control over real-time operations.

Inter-layer communication occurs only at harmonic cycle boundaries, ensuring timing consistency and preventing fault propagation across levels. This architecture embodies the CDEQ philosophy of hierarchical governance and fault tolerance.

9. Conclusion

CDEQ V10.0 establishes a foundational, provably stable control paradigm for rotating detonation engines, addressing the core engineering bottlenecks of overpressure, modal instability, and time-scale mismatch. By leveraging cross-system topological isomorphism, the framework inherits a century of mature diesel engine control logic, avoiding the "wave-chasing" trap that plagues conventional RDE control research.

Rooted in the five-element thermodynamic completeness theorem, the framework reduces the complexity of RDE dynamics to a minimal, controllable state space, with entropy and internal energy computable from directly measurable variables. The CDEQ Time Charter's layered governance principle—speed-governing-speed via FPGA hardware and slow-governing-fast via supervisory control—resolves the fundamental time-scale mismatch, ensuring deterministic control without disrupting detonation's natural rhythm.

Supported by the PCTT-CW topology, CENHE-LEX constraints, and GCVT-10 convergence interlock, the framework is designed for FPGA implementation in Q14 fixed-point arithmetic, complying with the strictest aerospace functional safety standards. This paper adheres to rigorous academic principles, providing a complete theoretical architecture and reproducible verification protocol without claiming experimental validation.

The CDEQ V10.0 framework does not attempt to "fight" the chaos of RDE dynamics; instead, it constrains and guides them using universal thermodynamic invariants and rigid time-scale discipline. It represents a paradigm shift in RDE control—from model-based intervention to topology-based constraint—providing a viable path for the engineering maturation of rotating detonation engines and their deployment in future hypersonic propulsion systems.

Acknowledgments

The author wishes to acknowledge the foundational contributions of Aleksandr Lyapunov, whose stability theory forms the mathematical bedrock of this framework. Gratitude is also extended to the pioneering researchers in the rotating detonation engine field, including Bykovskii, Wolanski, Frolov, and Paxson, whose work established the physical and experimental context for this control paradigm. This research received no external funding and represents independent theoretical and architectural development.

Intellectual Property Statement

All theoretical frameworks, formulas, algorithms, codes, and structural designs in this paper are the sole and original

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The Dao of CDEQ

Heart as Compass, Sword of Readiness.
Seek no trouble, fear no trouble.
Sword in sheath, hand on hilt.
Heart as anchor, mouth without noise.
Ignore us now.
The gate was never open.